1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
ERROR 1241 (21000): Operand should contain 1 column(s)
D2_96583_rina@localhost SELECT * FROM orders WHERE cnum = (SELECT cnum FROM
customers WHERE cname = 'Cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.012 sec)

D2_96583_rina@localhost
```

7. Extract all the orders of Motika.

```
D2_96583_rina@localhost SELECT * FROM orders where snum = (SELECT snum FROM
salespeople WHERE sname = 'MOTIKA');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.010 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
D2_96583_rina@localhost SELECT * FROM orders WHERE snum = ALL(SELECT snum FR
OM customers WHERE city = 'LONDON');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.014 sec)

D2_96583_rina@localhost
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
D2_96583_rina@localhost SELECT * FROM orders WHERE amt > ANY(SELECT amt  FRO
M orders o INNER JOIN customers c ON c.cnum = o.cnum WHERE city = 'LONDON');

+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.019 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
D2_96583_rina@localhost SELECT * FROM customers WHERE rating >= ANY(SELECT r
ating from customers c INNER JOIN salespeople s ON c.snum = s.snum WHERE sna
me = 'Serres');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.015 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
D2_96583_rina@localhost SELECT * FROM customers WHERE rating > ANY(SELECT ra
ting FROM customers WHERE city = 'ROME');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.013 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
D2_96583_rina@localhost SELECT * FROM orders WHERE amt > ANY(SELECT amt FROM
 orders WHERE odate = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.402 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
D2_96583_rina@localhost SELECT * FROM orders WHERE amt < ANY(SELECT amt FROM
 orders o INNER JOIN customers c ON o.cnum = c.cnum WHERE city = 'San Jose')
;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.188 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
D2_96583_rina@localhost SELECT o.amt, c.cname, c.rating FROM customers c INN
ER JOIN orders o ON c.cnum = o.cnum WHERE o.amt > ALL(SELECT AVG(amt) FROM o
rders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.508 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
3 rows in set (0.508 sec)

D2_96583_rina@localhost SELECT * FROM customers WHERE rating > ALL(SELECT ra
ting FROM customers WHERE city = 'LONDON');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.013 sec)

D2_96583_rina@localhost
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
D2_96583_rina>SELECT * FROM salespeople S  WHERE city != ALL(SELECT CITY fro
m customers);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
2 rows in set (0.647 sec)

D2_96583_rina>
```

9. Find names and numbers of all salesperson who have more than one customer.

```
D2_96583_rina@localhost SELECT snum, sname FROM salespeople WHERE snum = ANY
(SELECT snum FROM customers  GROUP BY snum HAVING COUNT(cnum) >1);
+------+--------+
| snum | sname  |
+------+--------+
| 1001 | Peel   |
| 1002 | Serres |
+------+--------+
2 rows in set (0.311 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
D2_96583_rina>SELECT s.snum, s.sname, s.city, (SELECT COUNT(*) FROM customer
s c WHERE c.snum = s.snum) AS customer_count FROM salespeople s WHERE (SELEC
T COUNT(*) FROM customers c WHERE c.snum = s.snum)>1;
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.531 sec)

D2_96583_rina>
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
D2_96583_rina@localhost SELECT snum, SUM(amt) AS TOTAL_AMT FROM orders GROUP
 BY snum HAVING SUM(amt)>(SELECT MAX(amt) FROM orders);
+------+-----------+
| snum | TOTAL_AMT |
+------+-----------+
| 1001 |  15382.07 |
+------+-----------+
1 row in set (0.017 sec)

D2_96583_rina@localhost
```